ZHENKAI GAO

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EDUCATION

Tufts University, Medford, MA Expected Dec 2027

M.S. in Mechanical Engineering - Human-Robot Interaction

Relevant Coursework: Human-Robot Interaction, Reinforcement Learning, Probabilistic Systems, Robotics, Statistical Machine Learning, Optimal Control

University of California, Davis, Davis, CA Sep 2023

B.S. in Mechanical Engineering

RESEARCH EXPERIENCE

Assistive Agent Behavior and Learning Lab (AABL), Tufts University Aug 2026 - Present

Graduate Researcher - Kinova Gen3 Lite Robot Learning & Shared Autonomy

Python | ROS 2 | MuJoCo | MJLab | CUDA | PPO

- Developed a goal-conditioned RL policy for a Kinova Gen3 Lite arm to reach multiple pre-grasp poses in a tea-serving task from randomized initial configurations.
- Built GPU-accelerated MuJoCo/MJLab environments and trained PPO policies, iterating on reward design and rollout evaluation to improve reaching performance.
- Integrated the learned policy with the physical Kinova Gen3 Lite through ROS 2 interfaces for sim-to-real deployment, achieving a 98% reaching success rate across randomized initial configurations.
- Collaborated with a PhD researcher to integrate and evaluate the policy within a shared-autonomy framework; co-authored a manuscript submitted to ACM/IEEE HRI.

Cyber-Human-Physical Systems (CHPS) Lab, University of California, Davis Jun 2022 - Jun 2023

Undergraduate Researcher

SolidWorks | ROS | Sensor Integration | Prototyping

- Designed and iterated the mechanical packaging of a compact handheld sensing platform integrating a thermal camera, IMU, and LiDAR for 3D mapping in unstructured and low-light environments.
- Supported hardware prototyping, sensor integration, assembly, soldering, and experimental system development.
- Co-authored a peer-reviewed paper on handheld sensor fusion for 3D mapping, published in *Sensors* (2024).

PROJECT EXPERIENCE

UR3e Grocery Pick-and-Place Jan 2026 - May 2026

Python | C++ | ROS 2 | RealSense D455 | YOLOv11 | TF2 | MoveIt 2

- Developed an end-to-end ROS 2 grocery pick-and-place system using a UR3e 6-DoF manipulator and RealSense D455, validated in simulation and on the physical robot.
- Integrated YOLOv11 RGB-D perception, TF2 coordinate transforms, and MoveIt 2 motion planning to generate pre-grasp, grasp, and lift motions from detected object locations.
- Implemented ROS 2 action clients, robot-control nodes, and a task state machine to coordinate perception, motion planning, arm execution, and gripper control.
- Designed and fabricated a 3D-printed TPU gripper, implemented serial gripper control, and configured wrist-mounted camera transforms for perception-to-robot alignment.

LEGO Signature Imitation Learning Robot Summer 2026

Python | MuJoCo | Imitation Learning | PPO | Computer Vision | Pure Pursuit

- Developed a differential-drive LEGO robot to reproduce human-drawn signatures using overhead vision and onboard IMU feedback for closed-loop pose estimation and trajectory tracking.
- Implemented a Pure Pursuit controller and trained an imitation-learning policy from its demonstrations for signature reproduction.
- Built a MuJoCo simulation environment and trained a PPO policy, incorporating system identification and calibration for sim-to-real deployment.
- Evaluated and compared Pure Pursuit control, imitation learning, and reinforcement learning on the physical robot using signature trajectory-tracking performance.

Bipedal Robot - Graduate Course Project Jan 2026 - May 2026

- Designed and 3D-printed mechanical components for a custom bipedal robot, iterating the design through gait testing.

SKILLS

Programming: Python, C++, MATLAB

Robotics & Simulation: ROS 2, MoveIt 2, TF2, RViz, Gazebo, MuJoCo, MJLab

Machine Learning & Perception: PyTorch, scikit-learn, Reinforcement Learning, OpenCV, YOLO

Mechanical Design: SolidWorks, Onshape, CAD, 3D Printing, Rapid Prototyping

Tools: Git, Linux, CUDA, CMake